

Music Mental Health Analysis

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Data

Data Background

The data used in this analysis comes from the Music and Mental Health (MxMH) Survey Results hosted on Kaggle. The dataset was collected through an online Google Form that was distributed via social media and public forums. It contains 736 respondents (rows) and includes variables on demographic characteristics (such as age), music-listening habits (hours per day, streaming platform, frequency of listening to certain genres), and self-reported mental-health scores for anxiety, depression, insomnia, and OCD (rated 0-10). The data is cross-sectional and self-reported, meaning it is suitable for identifying associations/correlations but not establishing causal effects.

Data Cleaning

Our data cleaning process began with an assessment of missing values across all columns. This revealed 107 null entries in the BPM column and a single null entry in the Age column. After looking further into the BPM data collection, we discovered that it was meant to represent the “Beats per minute of favorite genre.” However, rows with the same reported favorite genre displayed inconsistent BPM values, indicating that participants did not interpret or answer the question reliably. Combined with the large number of missing entries, this inconsistency made the variable unsuitable for analysis, so we dropped the BPM column entirely. The single row with a missing Age value was also removed to maintain completeness for demographic modeling.

Next, we examined the dataset for implausible or erroneous entries. During this process, we identified a respondent who reported being 89 years old and listening to music for 24 hours per day, and they were doing so while working. Because this combination is not physically

possible and would exert undue influence on any model involving hours per day, we removed this outlier from the dataset.

We then reviewed all variables designed to accept only binary “Yes” or “No” responses. Several of these columns contained values outside of the allowed set, such as misspellings, alternative phrasing, or blank entries. To maintain consistency in these binary predictors, we filtered out any rows that did not provide valid “Yes” or “No” answers across all such variables, resulting in the removal of nine additional rows.

After this, we checked the music effects variable and found seven cases where respondents left this question blank. Because an empty string does not constitute a valid category and this variable is used in later modeling, we removed these seven rows as well.

For analyses involving streaming services, we examined the primary streaming service column and identified two types of unusable entries: blank responses and the explicit answer “I do not use a streaming service.” Because our first research question specifically compares listening patterns across streaming platforms, respondents without a platform could not be meaningfully included in that analysis. Therefore, for the dataset used in RQ1, we removed all rows with missing streaming-service information as well as those who reported not using any streaming service.

Finally, to prepare the binary outcome for our second research question, we created a high depression variable that flags individuals with elevated symptom levels. Because the average depression score in the dataset was 4.82 and the median was 5, we designated respondents with a score of 6 or higher as experiencing “high depression.” This threshold represents a meaningful distinction between typical responses and substantially elevated symptom reports, allowing us to model the likelihood of more severe depression symptoms.

Models

Research Question 1: “How does the number of hours spent listening to music per day vary by depression status and primary streaming service?”

To investigate how the number of hours spent listening to music per day varies based on depression symptoms and primary streaming service, we fit a multiple linear regression model using hours per day as the continuous response variable. Linear regression is appropriate for this question because the goal is to estimate how the average listening duration changes across different levels of depression and across streaming platforms, while adjusting for potential confounders. We also included an interaction term between depression and primary streaming service, which tests whether the association between self-reported depression symptoms and daily listening time differs by the platform a respondent uses.

Beyond the primary predictors of depression and primary streaming service, we included a set of covariates to reduce potential confounding and isolate the effects of depression and

streaming service on listening time. These covariates were selected because they relate to both music-listening behavior and mental-health-related characteristics without making the model too busy. These covariates include age, while working (whether the respondent listens while working), instrumentalist (yes/no), composer (yes/no), exploratory (whether they tend to explore new music), foreign languages (whether they tend to listen to music in foreign languages), and three mental health related variables (anxiety, insomnia, OCD, all self-reported on a scale of 0-10). We also included two music-related behavioral factors, favorite genre and music effects, to account for genre preferences and how individuals perceive music impacts their emotions. Including these variables helps adjust for individual differences in music engagement, mental-health comorbidity, and daily listening contexts that could otherwise bias the estimated relationships.

Our model, which follows the linear regression framework, relies on the standard assumptions of linearity, independence of observations, homoscedasticity, and normally distributed residuals. We also assumed that there was no multicollinearity among predictors, and all of these assumptions were assessed via diagnostic plots/tests.

Research Question 2: “Does the number of hours spent listening to music each day increase the likelihood of reporting high depression symptoms?”

For our second research question, which sought to determine if the number of hours spent listening to music each day increases the likelihood of reporting high depression symptoms, we fit a logistic regression model with a binary outcome. Logistic regression is appropriate because the response variable, high depression, classifies respondents as either above or below a threshold level of self-reported depressive symptoms, and the model estimates how the log-odds of being in the high-depression category change with daily listening time.

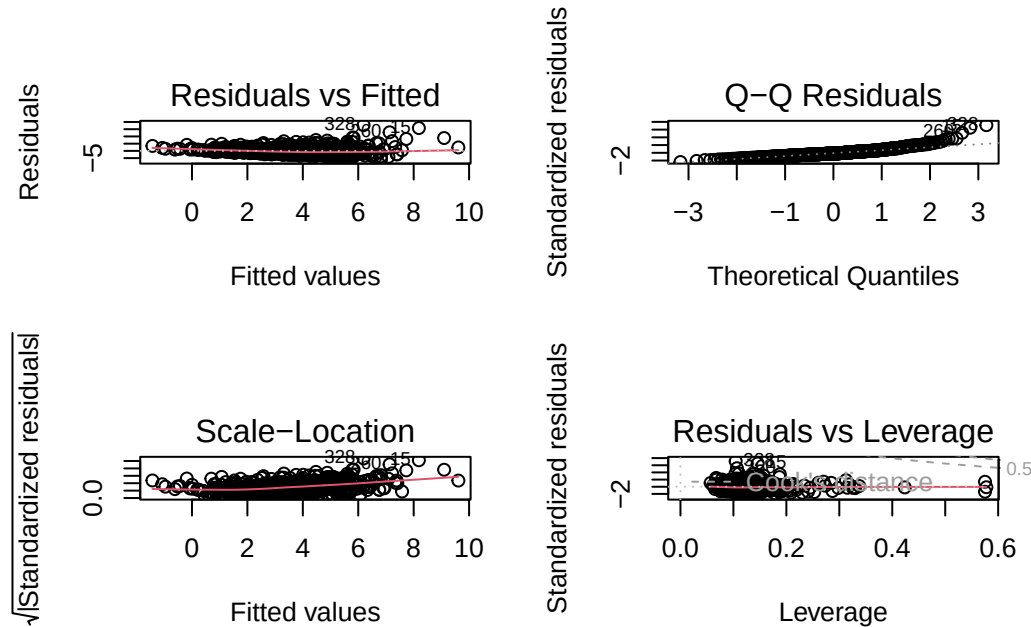
Outside of our outcome variable of high depression and our primary independent variable, daily hours spent listening to music, we also included several covariates. Specifically, we adjusted for age, indicators of music engagement (instrumentalist and composer), co-occurring mental-health conditions (anxiety, insomnia, and OCD), and two global music-use characteristics (favorite genre and music effects). These variables may influence both how much an individual listens to music and their underlying likelihood of experiencing elevated depressive symptoms, thereby helping reduce confounding effects.

Logistic regression, and therefore our model, assumes independent observations, linearity of continuous predictors on the log-odds scale, and the absence of multicollinearity, all of which were checked via plots and tests.

Assessment and changes

Research Question 1

The first model we fit for RQ1 was a linear regression model that regressed the raw outcome of hours of music listening per day on all available predictors in the data set.

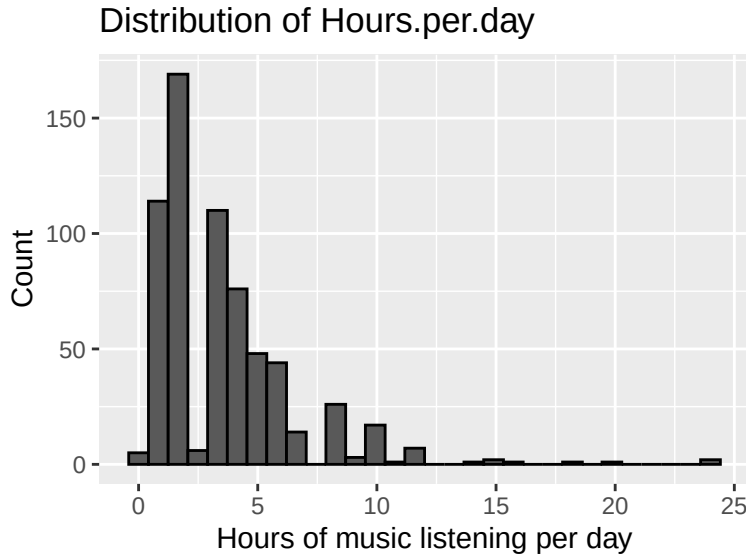


After fitting this initial model, we first examined the standard diagnostic plots, as shown above: Residuals vs Fitted, Normal Q-Q, Scale-Location, and Residuals vs. Leverage. The Residuals vs Fitted plot revealed a strong violation of homoscedasticity, with the points forming a funnel-shape, and a small violation of linearity, where the red smoothing line curved upward rather than remaining flat. The Normal Q-Q plot showed noticeable departures from the 45 degree line, especially in the upper tail, a clear strong violation of normality. The Scale-Location plot further confirmed heteroscedasticity, with residual spread increasing at higher fitted values. Finally, the Residuals vs. Leverage plot identified several influential observations with large residuals and unusually high leverage, suggesting they were exerting disproportionate influence on the fitted model.

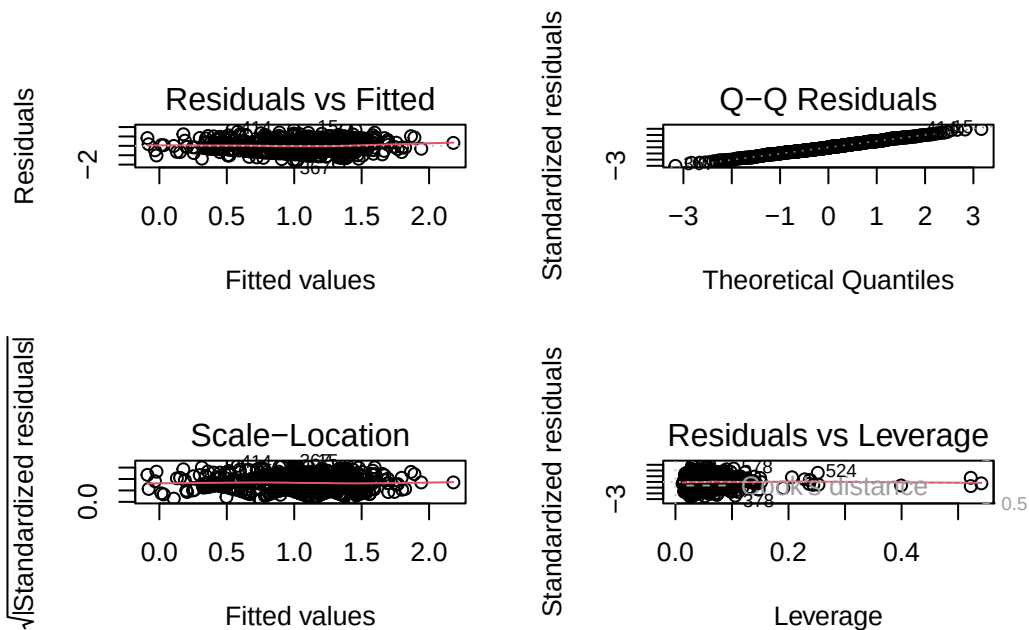
After examining the diagnostic plots, we assessed multicollinearity using GVIF values, which adjust for the degrees of freedom of multi-level categorical variables. For most predictors, including age, musician traits, mental-health indicators, and the genre-frequency variables, the adjusted $GVIF^{1/(2 \cdot Df)}$ values were all close to 1.1-1.3, indicating low multicollinearity and no immediate concern for inflation of variance. The interaction term between depression and

primary streaming service also showed a modest adjusted GVIF value of approximately 1.10, further supporting stability in these estimates.

One exception was favorite genre, a 16-level categorical variable, which produced a very large raw GVIF (2400). However, its adjusted value, $GVIF^{1/(2 \cdot Df)} \approx 1.30$, fell well within acceptable thresholds, since high raw GVIF numbers are expected for factors with many degrees of freedom. No predictors exhibited adjusted GVIF values approaching levels (2 or 3) that would indicate problematic multicollinearity.

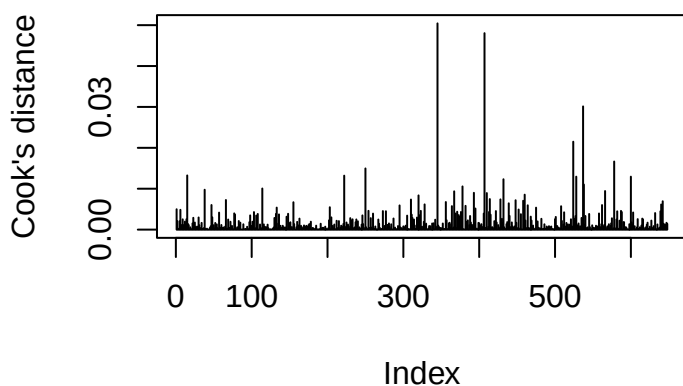


Because the full model included many weakly justified predictors, we fit a reduced model to improve parsimony and check whether the diagnostic issues were caused by unnecessary complexity. We also inspected a histogram of hours per day, pictured above, which was strongly right-skewed, suggesting structural issues with the outcome itself. The reduced model produced essentially the same adjusted R^2 (0.16) and similar residual behavior, with persistent violations of linearity, homoscedasticity, and normality, plus influential points. This showed that removing predictors did not improve model fit and that the underlying outcome distribution, not just model complexity, was driving the diagnostic problems.



To address this we applied a log transformation to the outcome and re-fit the model with $\log(\text{Hours per day} + 0.1)$ as the response. On the log scale, the Residuals vs fitted plot and Q-Q plot showed substantially more symmetric and homoscedastic residuals (as shown above) and the histogram of the transformed outcome indicated a much more balanced distribution. These improvements suggested that the log-transformed model better met the assumptions of linear regression. We also used variance inflation factors (VIFs) to check for multicollinearity among predictors and did not find strong evidence of a violation.

Influential points measured by Cook's distance



To evaluate influential observations, we computed Cook’s distance for the log-transformed RQ1 model and examined the corresponding plot. We also inspected the residual and leverage diagnostic plots for outliers. Combining these approaches, we identified three extreme points to remove: two points were flagged as outliers in both the diagnostic plots and Cook’s distance, and one additional point appeared as an outlier only in the diagnostic plots. After removing these three extreme points, we refit the log-transformed model. The residuals are now more symmetrically distributed (Min = -1.84, Max = 1.80, Median = 0), and the residual standard error decreased to 0.625, indicating improved model fit.

Several predictors remain significant, including Depression, Primary streaming service, While working, Instrumentalist, Composer, Insomnia, and select favorite genres, with interactions between Depression and streaming platform also holding. The adjusted R^2 increased to 0.220, showing that the log transformation and removal of influential points slightly improved the explained variance.

Table 1: Table 1. Model comparison for RQ1 using adjusted R-squared and AIC.

Model	Adj_R2	AIC
Raw model	0.1938095	3173.908
Reduced model	0.1579996	3158.997
Log transform	0.2324734	1309.501
Log transform + remove outliers	0.2198529	1260.190

Table 1 presents a comparison of the models for RQ1 using adjusted R^2 and AIC. The raw model explained about 19% of the variance but had a high AIC, reflecting poorer fit. Reducing predictors slightly lowered adjusted R^2 but improved AIC marginally. Applying the log transformation improved both adjusted R^2 (to 0.232) and AIC (to 1309), indicating better model performance. Finally, removing the three outliers slightly reduced adjusted R^2 to 0.220 but further lowered AIC to 1260, suggesting a more parsimonious and stable model. Overall, the log-transformed model with outliers removed balances interpretability, fit, and the influence of extreme points, and became our primary model for question 1.

Table 2: Table 2: Final linear regression model for RQ1 (log-hours with influential points removed).

Term	Estimate	Std Error	statistic	p-value
\(Intercept\)	-0.133	0.237	-0.5611308	0.575
Depression	0.106	0.032	3.2931047	0.001
Primary.streaming.serviceOther streaming service	0.696	0.245	2.8455457	0.005
Primary.streaming.servicePandora	0.776	0.355	2.1878891	0.029
Primary.streaming.serviceSpotify	0.577	0.197	2.9249186	0.004
Primary.streaming.serviceYouTube Music	0.625	0.216	2.8988409	0.004
Age	-0.005	0.003	-2.0325505	0.043

While.workingYes	0.605	0.065	9.3626702	<0.001
InstrumentalistYes	-0.160	0.063	-2.5549132	0.011
ComposerYes	0.152	0.075	2.0181945	0.044
ExploratoryYes	0.072	0.061	1.1673916	0.244
Foreign.languagesYes	0.060	0.054	1.1261125	0.261
Anxiety	0.000	0.011	0.0427548	0.966
Insomnia	0.026	0.009	2.8836975	0.004
OCD	0.019	0.010	1.9530351	0.051
Fav.genreCountry	0.149	0.168	0.8892265	0.374
Fav.genreEDM	0.248	0.149	1.6591055	0.098
Fav.genreFolk	0.094	0.161	0.5845425	0.559
Fav.genreGospel	0.060	0.317	0.1889684	0.850
Fav.genreHip hop	0.134	0.153	0.8818361	0.378
Fav.genreJazz	0.403	0.175	2.3074706	0.021
Fav.genreK pop	-0.072	0.175	-0.4120588	0.680
Fav.genreLatin	0.954	0.457	2.0872769	0.037
Fav.genreLofi	0.060	0.227	0.2636701	0.792
Fav.genreMetal	0.132	0.126	1.0502311	0.294
Fav.genrePop	-0.117	0.120	-0.9754466	0.330
Fav.genreR and B	0.128	0.153	0.8348958	0.404
Fav.genreRap	0.283	0.176	1.6140922	0.107
Fav.genreRock	0.097	0.115	0.8419446	0.400
Fav.genreVideo game music	-0.266	0.156	-1.7049786	0.089
Music.effectsNo effect	0.017	0.063	0.2704797	0.787
Music.effectsWorsen	-0.106	0.175	-0.6037773	0.546
Depression:Primary.streaming.serviceOther streaming service	-0.165	0.043	-3.8441528	<0.001
Depression:Primary.streaming.servicePandora	-0.207	0.067	-3.1180379	0.002
Depression:Primary.streaming.serviceSpotify	-0.103	0.033	-3.0904123	0.002
Depression:Primary.streaming.serviceYouTube Music	-0.140	0.038	-3.6613614	<0.001

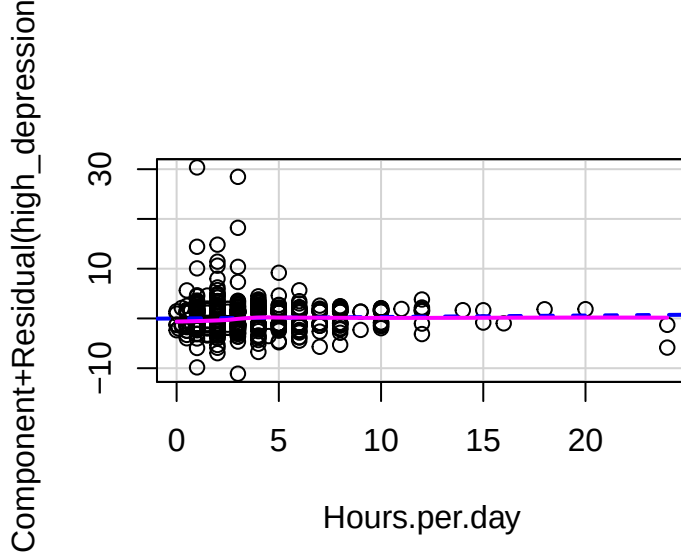
Table 2 shows that depression is associated with higher listening time ($\hat{\beta} = 0.106, p = 0.001$), but this effect is reduced for users of major streaming platforms, as indicated by the negative interaction terms (e.g., Spotify: $\hat{\beta} = -0.103, p = 0.002$; YouTube Music: $\hat{\beta} = -0.140, p < 0.001$). Several platforms also have positive main effects: Spotify ($\hat{\beta} = 0.577, p = 0.004$), YouTube Music ($\hat{\beta} = 0.625, p = 0.004$), Pandora ($\hat{\beta} = 0.776, p = 0.029$), and Other services ($\hat{\beta} = 0.696, p = 0.005$). All of these predict higher log-hours relative to the baseline streaming category, Apple Music.

Listening while working has the largest coefficient in the model ($\hat{\beta} = 0.605, p < 0.001$), indicating substantially higher daily listening. Additional significant predictors include insomnia

($\hat{\beta} = 0.026, p = 0.004$), being a composer ($\hat{\beta} = 0.152, p = 0.044$), being an instrumentalist ($\hat{\beta} = -0.160, p = 0.011$), age ($\hat{\beta} = -0.005, p = 0.043$), and two favorite genres: Jazz ($\hat{\beta} = 0.403, p = 0.021$) and Latin ($\hat{\beta} = 0.954, p = 0.037$). Most remaining predictors have confidence intervals crossing zero, indicating no detectable association with listening time.

Research Question 2

For RQ2, we modeled a binary outcome indicating “high depression” (depression score ≥ 6) using logistic regression, with daily listening time (Hours per day) as our primary predictor.



Because logistic regression assumes that continuous predictors have a linear relationship with the log-odds of the outcome, we began by checking this assumption for Hours per day using a component-plus-residual (partial residual) plot. This plot did not reveal strong curvature or systematic departures from linearity, as the pink line was fitted very tightly to the blue, so we retained Hours per day in linear form.

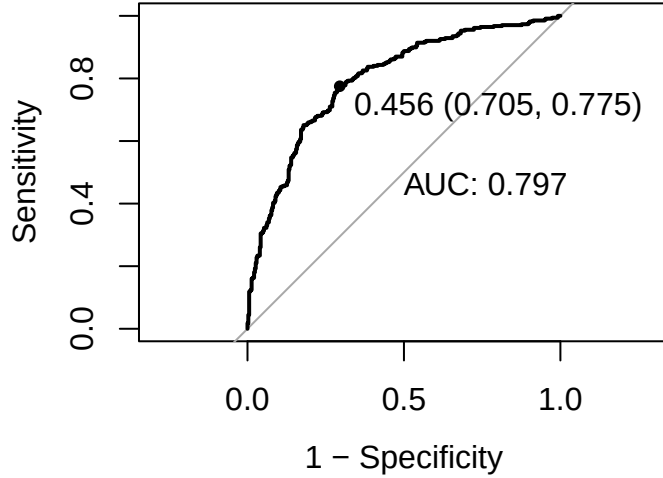
We then examined several additional diagnostics. First, we used VIFs to assess multicollinearity among the predictors and did not detect problematic levels of collinearity. Second, we evaluated overall model fit with a Hosmer–Lemeshow goodness-of-fit test, which revealed a p-value greater than 0.05 (0.1641), meaning the test indicates no substantial issue with the fitting of the model.

We calculated leverage values using the hat matrix and identified several observations above the common threshold of $2\bar{h}$. However, high leverage alone is not sufficient evidence to remove points in logistic regression, particularly in models with many categorical predictors. Because these cases did not show extreme Pearson residuals either, we retained all observations for the final model.

Table 3: Table 3: Confusion Matrix and Classification Metrics

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced A
277	95	103	243	0.724	0.447	0.719	0.729	0.702	0.719	0.711	

To assess predictive performance, we computed an initial confusion matrix (Table 3) based on the conventional 0.5 probability threshold. The initial confusion matrix showed 277 true negatives, 243 true positives, 95 false negatives, and 103 false positives, yielding an overall accuracy of 0.724. Sensitivity was 0.719, meaning the model correctly identified about 72% of participants with high depression, while specificity was 0.729, indicating a similar rate of correct classification among low-depression participants. The model’s precision (0.702) and F1 score (0.711) suggested moderate predictive performance but also revealed room for improvement, particularly in reducing false negatives.



To further assess model performance, we examined the ROC curve and computed the AUC. The model achieved an AUC of 0.797, indicating strong discriminative ability. The optimal classification threshold was identified as 0.456, which balances sensitivity and specificity more effectively than the default 0.50 cutoff.

Table 4: Table 4: Confusion Matrix and Classification Metrics (Threshold = 0.456).

TN	FP	FN	TP	Accuracy	Kappa	Sensitivity	Specificity	Precision	Recall	F1	Balanced A
268	76	112	262	0.738	0.478	0.775	0.705	0.701	0.775	0.736	